

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Bayesian view of the LS Periodogram

- What we want to be able to do is to detect variability and measure the period in the face of both noisy and incomplete data. Instead we'll use Fourier decomposition to get a more useful tool for actual data analysis.
- For a periodic signal we have:

$$y(t + P) = y(t),$$

where P is the period.

- We can create a *phased light curve* that plots the data as function of phase:

$$\phi = \frac{t}{P} - \text{int} \left(\frac{t}{P} \right),$$

- where $\text{int}(x)$ returns the integer part of x .

A Single Sinusoid

- Let's take the case where the data are drawn from a single sinusoidal signal:

$$y(t) = A \sin(\omega t + \phi) + \epsilon$$

- and determine whether or not the data are indeed consistent with periodic variability and, if so, what is the period.
- This model is **non-linear** in the frequency term, ω and the phase, ϕ and therefore We rewrite the argument as $\omega(t - t_0)$ (reexpressing the phase term) and use trigonometric identities to rewrite the model as:

$$y(t) = a \sin(\omega t) + b \cos(\omega t)$$

- where

$$A = (a^2 + b^2)^{1/2} \text{ and } \phi = \tan^{-1}(b/a)$$

- The model is now linear with respect to coefficients a and b (and nonlinear only with respect to frequency, ω).

- Assuming constant uncertainties on the data, we can write a likelihood function down:

$$L = \prod_{j=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{[y_j - a \sin(\omega t_j) - b \cos(\omega t_j)]^2}{2\sigma^2} \right)$$

- where y_i is the measurement (e.g., the brightness of a star) taken at time t_i . - With a lot of math we do not report here, and assuming uniform priors on a, b, ω , and σ (which gives nonuniform priors on A and ϕ), the posterior distribution of parameters can be simplified to:

$$p(\omega, a, b, \sigma | t, y) \propto \sigma^{-N} \exp \left(\frac{-NQ}{2\sigma^2} \right)$$

- with

$$Q = V - \frac{2}{N} \left[a I(\omega) + b R(\omega) - a b M(\omega) - \frac{1}{2} a^2 S(\omega) - \frac{1}{2} b^2 C(\omega) \right]$$

- and

$$V = \frac{1}{N} \sum_{j=1}^N y_j^2$$

$$I(\omega) = \sum_{j=1}^N y_j \sin(\omega t_j)$$

$$R(\omega) = \sum_{j=1}^N y_j \cos(\omega t_j)$$

$$M(\omega) = \sum_{j=1}^N \sin(\omega t_j) \cos(\omega t_j)$$

$$S(\omega) = \sum_{j=1}^N \sin^2(\omega t_j)$$

$$C(\omega) = \sum_{j=1}^N \cos^2(\omega t_j)$$

- Note that I, R, M, S, C only depend on ω and the data.

- If $N \gg 1$ and we have data that extends longer than the period:

$$S(\omega) \approx C(\omega) \approx N/2 \text{ and } M(\omega) \ll N/2$$

- and

$$Q \approx V - \frac{2}{N} [a I(\omega) + b R(\omega)] + \frac{1}{2} (a^2 + b^2)$$

The posterior for many, randomly spaced, observations

- If we marginalize over $\mathbf{a}$ and $\mathbf{b}$ (as we are interested in the period):

$$p(\omega, \sigma | \{t, y\}) \propto \sigma^{-(N-2)} \exp \left(\frac{-NV}{2\sigma^2} + \frac{P(\omega)}{\sigma^2} \right)$$

- with

$$P(\omega) = \frac{1}{N} [I^2(\omega) + R^2(\omega)]$$

$$V = \frac{1}{N} \sum_{j=1}^N y_j^2$$

$$I(\omega) = \sum_{j=1}^N y_j \sin(\omega t_j)$$

$$R(\omega) = \sum_{j=1}^N y_j \cos(\omega t_j)$$

- we know the noise σ then

$$p(\omega | \{t, y\}, \sigma) \propto \exp \left(\frac{P(\omega)}{\sigma^2} \right)$$

- and we now have the posterior for ω !

Significance of the peaks in the periodogram

- Let's compute the χ^2 for the LS periodogram:

$$\chi^2(\omega) \equiv \frac{1}{\sigma^2} \sum_{j=1}^N [y_j - y(t_j)]^2 = \frac{1}{\sigma^2} \sum_{j=1}^N [y_j - a_0 \sin(\omega t_j) - b_0 \cos(\omega t_j)]^2$$

- which we can simplify to:

$$\chi^2(\omega) = \chi_0^2 \left[1 - \frac{2}{NV} P(\omega) \right]$$

- where, again, $P(\omega)$ is the periodogram and χ_0^2 is the χ^2 for a model with $y(t)=\text{constant}$:

$$\chi_0^2 = \frac{1}{\sigma^2} \sum_{j=1}^N y_j^2 = \frac{NV}{\sigma^2}$$

- We'll now renormalise the periodogram as:

$$P_{\text{LS}}(\omega) = \frac{2}{NV} P(\omega),$$

- where $0 \leq P_{LS}(\omega) \leq 1$.
- With this renormalization, the reduction in $\chi^2(\omega)$ for the harmonic model, relative to χ^2 for the pure noise model, χ_0^2 is:

$$\frac{\chi^2(\omega)}{\chi_0^2} = 1 - P_{LS}(\omega).$$

- To determine if our source is variable or not, we first compute $P_{LS}(\omega)$ and then model the odds ratio for our variability model vs. a no-variability model.
- If our variability model is "correct", then the peak of $P(\omega)$ gives the best ω and the χ^2 at $\omega = \omega_0$ is N .

- If the true frequency is ω_0 then the maximum peak in the periodogram should have a height:

$$P(\omega_0) = \frac{N}{4}(a_0^2 + b_0^2)$$

- and standard deviation:

$$\sigma_P(\omega_0) = \frac{\sqrt{2}}{2} \sigma^2.$$

Credits

Course Flow

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	Previous lecture	Next lecture	Course Summary
notebook	<u>Irregular sampling</u>	<u>Irregular sampling</u>	<u>Course Summary</u>
html	<u>Irregular sampling</u>	<u>Irregular sampling</u>	<u>Course Summary</u>

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